

20CS5T03 – Data Warehousing & Mining

UNIT 1: Data Warehousing Concepts

Introduction

Data warehousing involves collecting and managing large volumes of data.

Detailed Explanation

Characteristics

Subject-oriented, integrated, time-variant.

Architecture

Includes ETL process.

Applications

Used in business intelligence.

Summary

This unit explains data warehousing.

UNIT 2: Data Preprocessing

Introduction

Prepares data for mining.

Detailed Explanation

Data Cleaning

Removes noise.

Data Transformation

Converts formats.

Applications

Used in analytics.

Summary

This unit explains preprocessing.

UNIT 3: Data Mining Techniques

Introduction

Extracts patterns from data.

Detailed Explanation

Classification and Clustering

Identify patterns.

Applications

Used in analytics.

Summary

This unit explains mining.

UNIT 4: Association Rules

Introduction

Finds relationships.

Detailed Explanation

Apriori Algorithm

Finds frequent itemsets.

Applications

Used in market analysis.

Summary

This unit explains association.

UNIT 5: Clustering and Classification

Introduction

Groups data.

Detailed Explanation

Techniques

K-means, decision trees.

Applications

Used in ML.

Summary

This unit explains clustering.